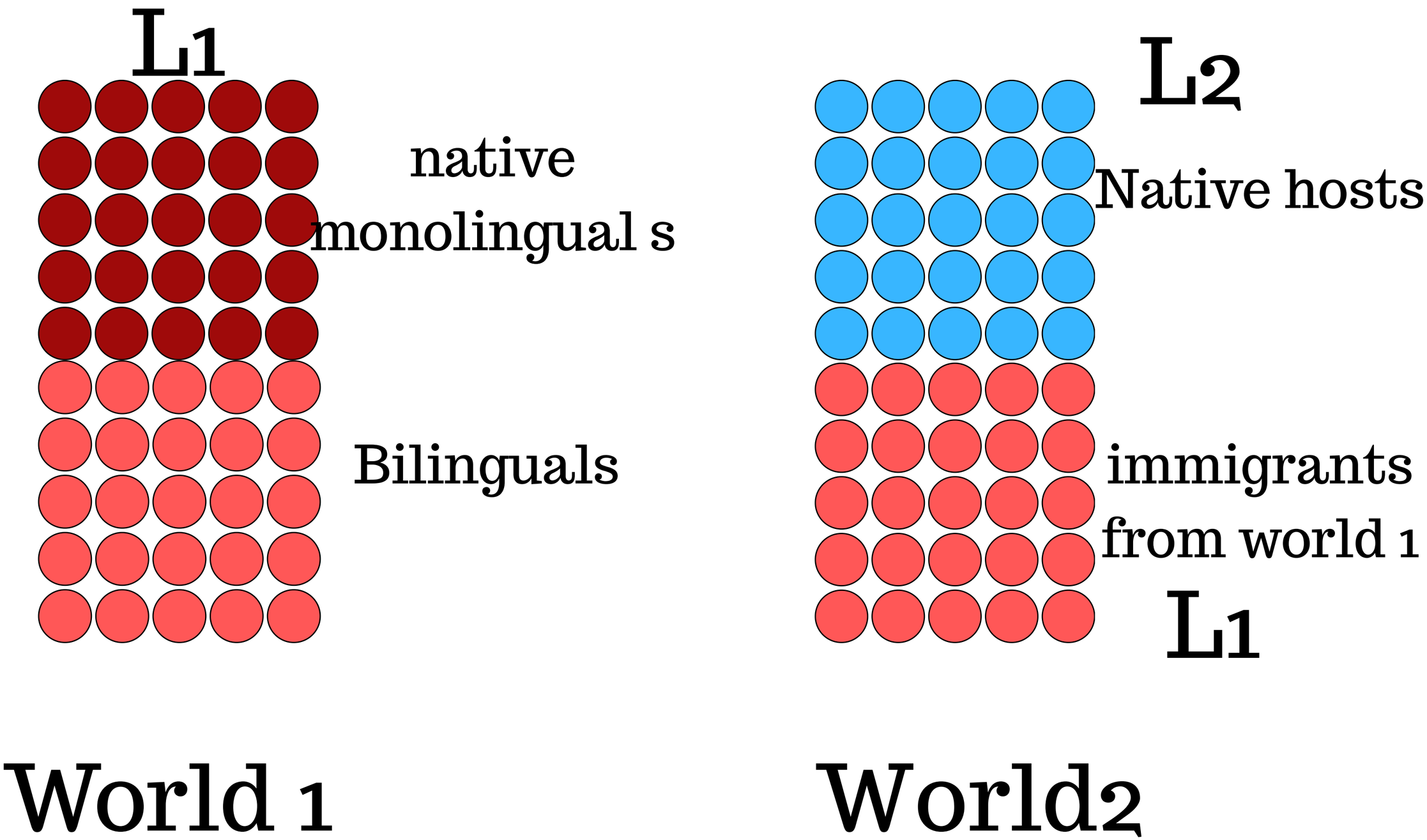
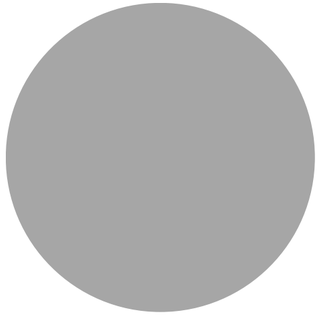
How does the communication success of color terms emerge through linguistic pressure modulated by geographical location

Meissa and Mike


L1
native
monolingual s
Bilinguals
World 1
L2
Native hosts
immigrants
from world 1
L1
World2

In world 1, agents have these respective lists:

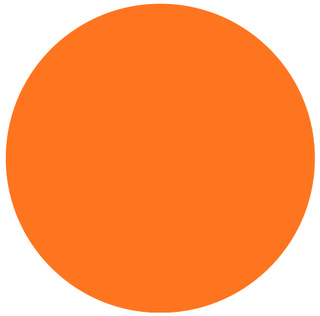
yellow



monolingual

A1 in Language 1

yellow
red
in Language 1



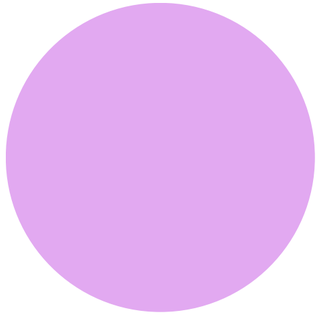
bilingual

A2

Jaune
rouge
in Language 2

In <u>world 2</u>, agents have these respective lists:

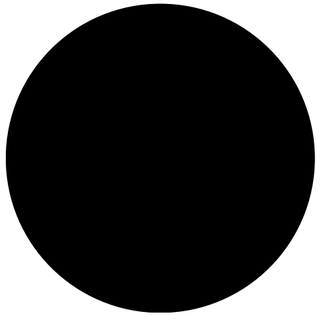
yellow



A1

in Language 2

yellow
red



A2

Jaune
rouge

in Language 1

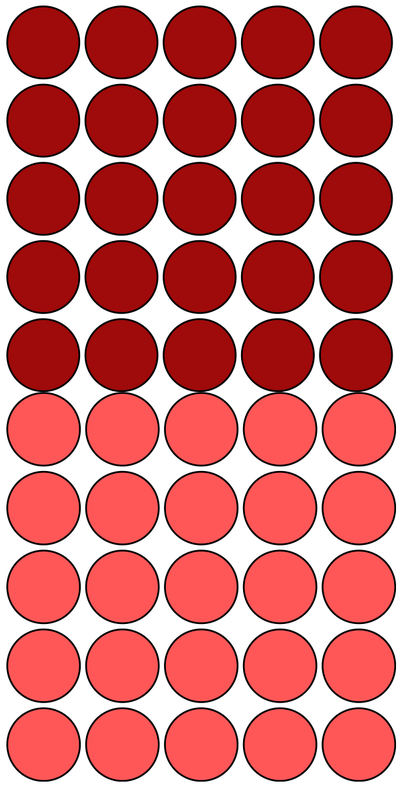
in Language 2

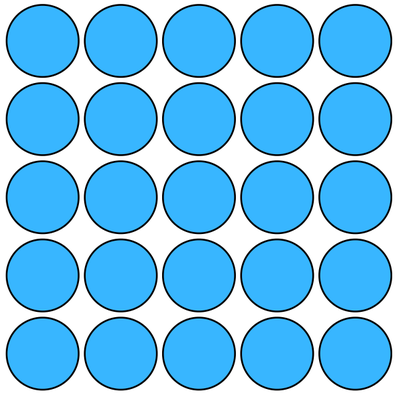
- Agents should increase their communication success rate by having their token weight increasing
- the number of monolingual agents should always be superior to bilinguals in both worlds (we can start with a 3/2 ratio :)
- communication success is modulated by the probability for an agent to meet another. Therefore, if the probability for a bilingual agent (A2) to talk to a monolingual agent (A1) is high, the token weight should increases. Same apply for a bilingual agent (A2) to talk to another bilingual.
- Agents will try to match their guesses when communicating color terms. If the matching is wrong the agent will find another peer to initiate another communication. On the contrary, if their guesses converge on the same meaning, the weight of the token increases.
- Eventually, agents will try to communicate only to agents that match most of their tokens weight
- Bilinguals in World1 will always use the Language 1 to communicate with monolingual agents.
- Bilinguals in World2 will communicate in both Language 1 and Language 2 with native hosts and immigrants.

First possible outcome:

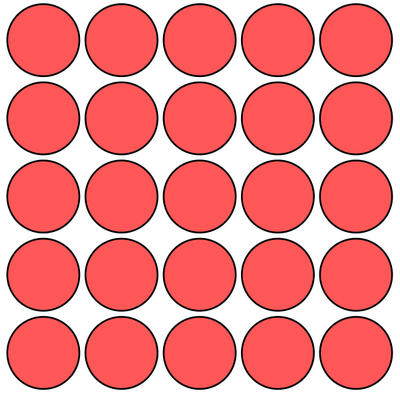
agents in Worl2 have lower matching rate in token weight in Language 1 with the native hosts. The communication success drops. Their social bonding is weak, and form two clusters, one indicate social exclusion of the immigrant agents (A2).

agents in Worl1 have greater matching rate in token weight, and higher communication success. Their social bonding is strong.



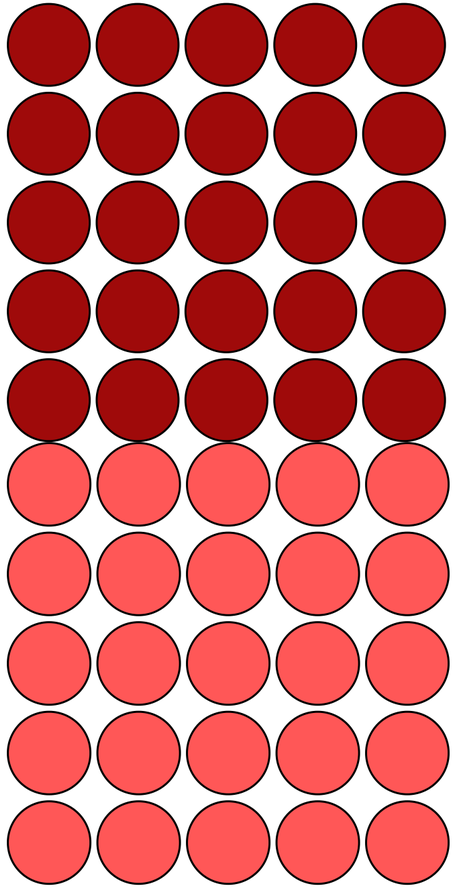


In language 1, immigrants or agent 2 start to form a “ghetto” group and their communication success increases between them because their token weight matching is identical.

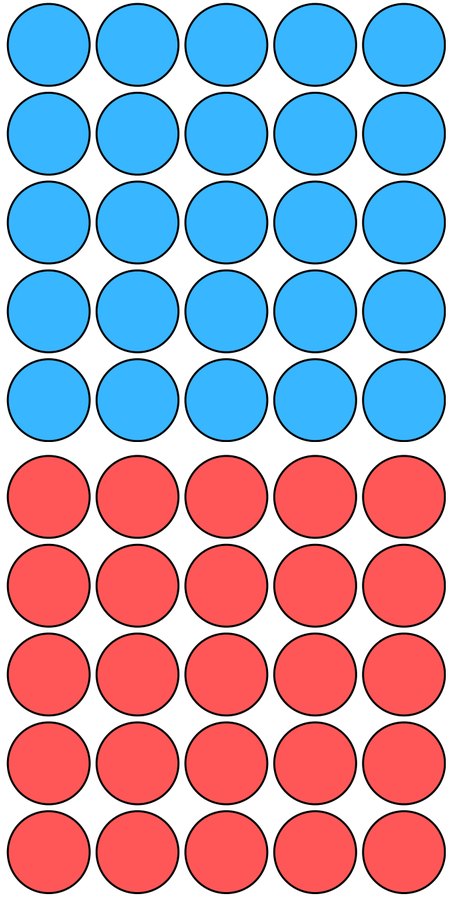


2nd possible outcome:

still bonding. nothing changes in World1.



Social bonding is strong in World 2, immigrants start to use token in Language 2 to increase communication success with native hosts.



Please add any rules or other suggestions to model the social interactions of the agents if you any in mind :)

Cheers!